

## Article

# A Hybrid Heuristic–Benders Method for Wind–Hydrogen Investment Planning with Non-Analytical Cost Functions

Haozhe Xiong <sup>1</sup>, Bingyang Feng <sup>1</sup> , Fangbin Yan <sup>1</sup>, Yiqun Kang <sup>1</sup>, Yuxuan Hu <sup>1</sup>, Qiangsheng Li <sup>2</sup> and Qinyue Tan <sup>2,\*</sup> 

<sup>1</sup> State Grid Hubei Electric Power Research Institute, State Grid Hubei Electric Power Co., Ltd., Wuhan 430077, China; xionghz\_sgcc@163.com (H.X.); fby6923@foxmail.com (B.F.); 790221951@qq.com (F.Y.); kangyq\_hbdky@163.com (Y.K.); karryhyx@163.com (Y.H.)

<sup>2</sup> College of Water Resources and Architectural Engineering, Northwest A&F University, Yangling 712100, China; liqiangsheng@nwfau.edu.cn

\* Correspondence: qinyuetan@nwsuaf.edu.cn

## Abstract

This paper studies capacity planning for a wind–hydrogen integrated energy system under scenario-based uncertainty in wind generation, hydrogen demand, and electricity prices. The model is formulated as a two-stage stochastic program in which first-stage investment decisions are selected before uncertainty is realized and second-stage hourly operation is optimized for each representative scenario. The main methodological difficulty is that part of the first-stage hydrogen-storage investment cost may be available only through a non-analytical evaluator, such as supplier quotation logic, simulation software, or a data-driven estimator, while the operational recourse model remains linear. To address this setting, a hybrid heuristic–Benders framework, denoted as GSOA-Benders, is developed by coupling the General-Soldiers Optimization Algorithm for derivative-free first-stage search with Benders cuts generated from linear programming subproblems. The framework is not presented as a replacement for commercial solvers on explicit convex or mixed-integer models; rather, it is intended for cases where exact algebraic reformulation of the first-stage cost is unreliable or unavailable. In the black-box case study with 500 scenarios, the method converges in 35.86 s and obtains an investment plan expressed as  $x = [1, 0.53, 23.23, 0]$ , corresponding to wind-farm construction, a 0.53 MW electrolyzer, a 23.23 MWh hydrogen tank, and no fuel-cell investment. Additional discussion is provided on stability-gap interpretation, benchmark limitations, component lifetime assumptions, hydrogen losses, and environmental extensions.

**Keywords:** wind–hydrogen integrated energy system; two-stage stochastic programming; Benders decomposition; black-box optimization; investment planning



Academic Editor: Tonio Sant

Received: 13 April 2026

Revised: 27 April 2026

Accepted: 29 April 2026

Published: 30 April 2026

**Copyright:** © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and

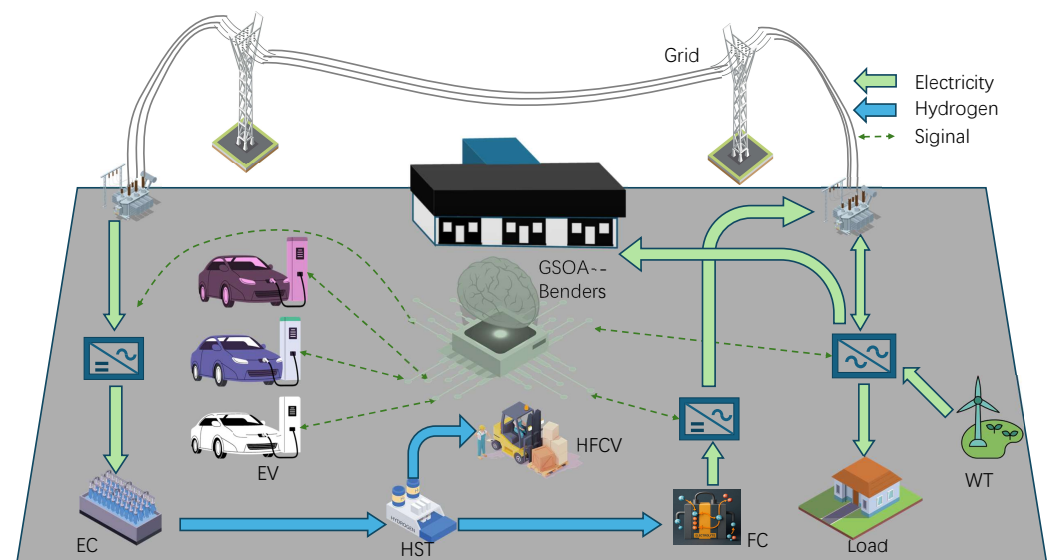
conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

## 1. Introduction

The global energy system is undergoing a profound transformation. To mitigate climate change and support deep decarbonization, renewable generation is expanding rapidly worldwide [1]. According to IRENA, global installed renewable power capacity reached 4443 GW at the end of 2024, including 1133 GW of wind power, while the IEA reports that global installed dedicated electrolyzer capacity reached about 1.4 GW by the end of 2023 [2,3]. These trends highlight the growing practical relevance of jointly planning renewable generation and hydrogen conversion infrastructure. However, the power output of renewable energy sources is characterized by intermittency, volatility, and uncertainty.

Consequently, their large-scale grid integration poses significant challenges to the real-time balance and operational security of power systems [4,5].

Hydrogen energy, particularly green hydrogen produced by electrolyzing water with surplus renewable energy, is regarded as a key enabler in addressing these challenges and bridging the power sector with other energy sectors (i.e., power to gas, P2G), as shown in Figure 1 [6,7]. In large power systems with fast-growing renewable penetration, such as provincial power grids in China, renewable accommodation increasingly requires the coordinated deployment of multiple storage technologies, including pumped storage, electrochemical batteries, gravity storage, and hydrogen storage. Compared with short-duration electrochemical storage, hydrogen can provide long-duration and cross-sector flexibility: surplus wind energy can be converted into hydrogen during off-peak hours; stored over longer periods; then consumed by industry, transport, or buildings or reconverted to electricity when economically justified. Therefore, hydrogen is not only an energy-storage carrier but also an energy-consumption pathway that links renewable electricity with hard-to-electrify end uses [8,9].



**Figure 1.** Overall schematic of the wind–hydrogen integrated energy system.

Consequently, the planning and design of Wind–Hydrogen Integrated Energy Systems (WH-IESs)—integrating wind power generation, electrolyzers, hydrogen storage tanks, and fuel cells—have emerged as a critical research focus globally [10–12]. WH-IES components require substantial initial investments, while their operational benefits depend on uncertain wind resources, demand profiles, and market electricity prices over the project horizon. Determining the trade-off between upfront investment and uncertain future operation is therefore a two-stage stochastic investment-planning problem.

Existing studies have developed several important methodological streams. First, stochastic and robust planning models have been widely used for renewable-dominant microgrids and integrated energy systems [13–15]. In particular, column-and-constraint generation (C&CG) has become a powerful approach for explicit two-stage or tri-level robust planning models. For example, recent work on bi-directional converter-based interconnection planning for renewable-dominant hybrid microgrids proposes a tri-level robust planning framework and a fully parallel C&CG algorithm to handle renewable uncertainty and converter efficiency effects [16]. Such studies demonstrate the strength of decomposition when the uncertain recourse structure is explicitly modeled and can be reformulated or convexified. Second, monolithic solvers and MILP/MINLP reformulations remain highly effective when all investment and operation functions can be expressed ana-

lytically [17–20]. Third, heuristic and surrogate-assisted optimization methods can search non-convex landscapes without derivatives and have been applied to hybrid renewable-energy planning [21–25].

Nevertheless, an important gap remains. In many engineering-planning tasks, part of the first-stage investment cost is not provided as a clean algebraic function. Examples include supplier quotation engines with hidden discount rules, site-specific civil-work costs from simulation software, safety-distance and land-acquisition adjustments, and trained machine learning estimators. These cost evaluators may be queried for a candidate capacity vector but cannot always be reformulated reliably as a compact MILP or MINLP. At the same time, the second-stage operational model is often linear and scenario-separable. This creates a mixed structure: a black-box first-stage cost coupled with an explicit large-scale linear recourse model.

The present paper addresses this specific structure. It does not claim that combining a metaheuristic with Benders decomposition is fundamentally new in itself. Rather, the contribution lies in formulating and testing a WH-IES planning framework in which (i) the non-analytical first-stage investment evaluator is isolated and searched by a derivative-free algorithm, (ii) the expected second-stage operating cost is approximated by valid Benders cuts derived from linear recourse duals, and (iii) the convergence indicator is interpreted as a stability gap rather than a strict global optimality certificate. This positioning distinguishes the proposed framework from explicit C&CG/Benders models, pure simulation-based heuristics, and surrogate-only approaches.

The main contributions of this paper are summarized as follows:

1. A hybrid GSOA-Benders framework is developed for a practically relevant class of WH-IES planning problems with black-box first-stage investment costs and scenario-based linear second-stage operation. The novelty is application-level and algorithmic integration, not the first use of heuristic decomposition in the literature.
2. The role and limitation of Benders cuts are clarified formally. The cuts provide valid lower approximations for the convex expected recourse function, whereas the non-convex black-box investment term is handled by a derivative-free master solver. Therefore, the reported convergence measure is defined as a stability gap and is not overstated as a rigorous global optimality gap.
3. A transparent case study compares the proposed framework with monolithic, reformulation-based, exhaustive simulation, and heuristic-decomposition benchmarks. The discussion is deliberately balanced: commercial solvers remain preferred for explicit models, while the proposed method is intended for cases where black-box first-stage costs prevent direct exact formulation.
4. The revised model discussion explicitly addresses practical modeling issues raised by reviewers, including hydrogen losses, component lifetime and degradation, the stylized nature of the sinusoidal tank-cost benchmark, scalability boundaries, and possible emission-reduction extensions.

The remainder of this paper is organized as follows. Section 2 presents the mathematical formulation and clarifies its relation to explicit C&CG/Benders approaches. Section 3 details the GSOA-Benders solution framework and the stability-gap definition. Section 4 presents the case study, benchmark comparisons, sensitivity discussion, and operational analysis. Section 5 concludes the paper.

## 2. Mathematical Formulation

The investment planning problem for the WH-IES investigated in this paper is formulated as a two-stage stochastic programming problem under uncertainty. In this paper, “stochastic” refers to the use of a finite scenario set with probabilities; the model is not

a classical two-stage robust optimization model with a worst-case uncertainty set. This clarification is important because some robust planning algorithms, including C&CG, solve max–min or tri-level robust counterparts, whereas the present paper minimizes the expected operating cost over representative scenarios.

### 2.1. Position Relative to C&CG and Classical Benders Decomposition

Classical Benders decomposition and C&CG methods are strongest when both the first-stage master problem and the second-stage recourse or adversarial subproblem can be written explicitly. For a generic two-stage robust model, C&CG alternates between a master problem containing accumulated worst-case constraints and a subproblem that identifies new worst-case realizations. This logic has been successfully applied to renewable-dominant microgrids, including the recent bi-directional-converter interconnection planning work in which converter-efficiency nonlinearity was convexified and solved within a fully parallel C&CG framework [16].

The present problem has a different computational bottleneck. The operational recourse model remains an LP and is therefore suitable for Benders cuts, but the first-stage investment cost may be available only through a black-box evaluator. If this evaluator is a compiled simulation, a supplier pricing service, or a trained neural network, an exact C&CG or MILP reformulation of the master problem may be unavailable. Therefore, the proposed framework keeps the Benders logic only for the explicit recourse function and replaces the exact master solver with the GSOA. This is the reason the method is described as a heuristic–Benders framework and why its convergence indicator is defined conservatively in Section 3 rather than being presented as a proof of global optimality.

### 2.2. Two-Stage Optimization Framework

The decision-making process is decomposed into two distinct stages:

1. Stage I (Investment Decisions): These represent the here-and-now decisions. The planner determines the optimal installed capacities of the equipment, denoted by the decision vector ( $\mathbf{x}$ ), prior to the realization of uncertainties (e.g., wind power availability and load demand).
2. Stage II (Operational Decisions): These correspond to the wait-and-see recourse decisions. Once the uncertainty ( $\xi$ , representing a specific operational scenario) is realized, the system operator determines the optimal operational strategy ( $\mathbf{y}$ ) based on the fixed capacity ( $\mathbf{x}$ ) under scenario  $\xi$ .

Uncertainty is characterized by a set of  $N$  discrete stochastic scenarios indexed by  $s \in \mathcal{S} = \{1, 2, \dots, N\}$ . In this case study,  $N$  is set to 500. Each scenario ( $s$ ) is associated with a probability of occurrence ( $\pi_s$ , where  $\pi_s = 1/N$ ).

The overarching objective is to minimize the aggregate cost, comprising the Stage I annualized investment cost and the Stage II expected annualized operational cost. The compact mathematical formulation is expressed in Equation (1):

$$\begin{aligned} \min_{\mathbf{x}} \quad \mathcal{F} &= C_{inv}(\mathbf{x}) + \mathbb{E}_{\xi}[Q(\mathbf{x}, \xi)] \\ &= C_{inv}(\mathbf{x}) + \sum_{s=1}^N \pi_s \cdot Q(\mathbf{x}, \xi_s) \end{aligned} \quad (1)$$

where  $C_{inv}(\mathbf{x})$  represents the investment cost function (Stage I) and  $Q(\mathbf{x}, \xi_s)$  denotes the optimal operational cost (Stage II) under scenario  $s$ , given the investment decision ( $\mathbf{x}$ ).

### 2.3. Master Problem (MP): Investment Decision Model

The Master Problem (MP) aims to determine the optimal investment vector ( $\mathbf{x}$ ) and the minimized expected total cost ( $\mathcal{F}$ ). The formulation is given by

$$\min_{\mathbf{x}, \vartheta} \mathcal{F}_{MP} = C_{inv}(\mathbf{x}) + \vartheta \quad (2)$$

$$\text{s.t. } \mathbf{x} \in \Omega \quad (3)$$

$$\vartheta \geq \vartheta^{\min} \quad (\text{Lower bound proxy}) \quad (4)$$

$$\text{Benders Cuts (Detailed in Section 3)} \quad (5)$$

(1) Decision Variables ( $\mathbf{x}$ ): The investment decision vector ( $\mathbf{x} = [x_{WT}, P_{EL}, E_{H2}, P_{FC}]^T$ ) comprises the following:

- $x_{WT} \in \{0, 1\}$ : Binary decision for wind farm construction (1 denotes construction; 0 otherwise);
- $P_{EL} \in \mathbb{R}^+$ : Installed capacity of the electrolyzer (MW);
- $E_{H2} \in \mathbb{R}^+$ : Capacity of the hydrogen storage tank (MWh);
- $P_{FC} \in \mathbb{R}^+$ : Installed capacity of the fuel cell (MW).

(2) Investment Cost Function: The annualized investment cost ( $C_{inv}(\mathbf{x})$ ) is calculated over a lifespan of  $\mathcal{L} = 10$  years:

$$C_{inv}(\mathbf{x}) = \frac{1}{\mathcal{L}} \left[ C_{WT}(x_{WT}) + C_{EL}(P_{EL}) + C_{FC}(P_{FC}) + C_{H2}(E_{H2}) \right] \quad (6)$$

The specific cost components are defined as follows:

#### 2.3.1. Wind Power Cost (Binary)

$$C_{WT}(x_{WT}) = \kappa_{WT} \cdot x_{WT} \quad (7)$$

where  $\kappa_{WT}$  represents the fixed turnkey cost (\$75 million) for a wind farm with a fixed capacity (e.g.,  $P_{WT}^{rated} = 50$  MW).

#### 2.3.2. Electrolyzer and Fuel-Cell Costs (Linear)

$$C_{EL}(P_{EL}) = \kappa_{EL} \cdot P_{EL}, \quad C_{FC}(P_{FC}) = \kappa_{FC} \cdot P_{FC} \quad (8)$$

where  $\kappa_{EL}$  and  $\kappa_{FC}$  denote the linear unit capacity costs (\$/MW).

#### 2.3.3. Hydrogen Tank Cost (Complex Non-Convex Model) [26]

The investment cost of the hydrogen storage tank, denoted as  $C_{H2}(E_{H2})$ , is modeled using a composite function that integrates a minimum investment threshold, economies of scale with discounted pricing tiers, and cyclic modularity costs [27]. As shown in Figure 2, this formulation renders the problem intractable for standard convex solvers:

$$C_{H2}(E_{H2}) = \max \left\{ C_{\min}, \quad C^{\text{base}}(E_{H2}) + C^{\text{cyc}}(E_{H2}) \right\} \quad (9)$$

where  $E_{H2}$  represents the rated capacity of the hydrogen tank (MWh). The function consists of two components:

- (1) Base Cost with Discount Scheme (Piecewise & Non-convex): Reflecting economies of scale, the base cost follows a power function (0.7). Furthermore, a step-wise discount

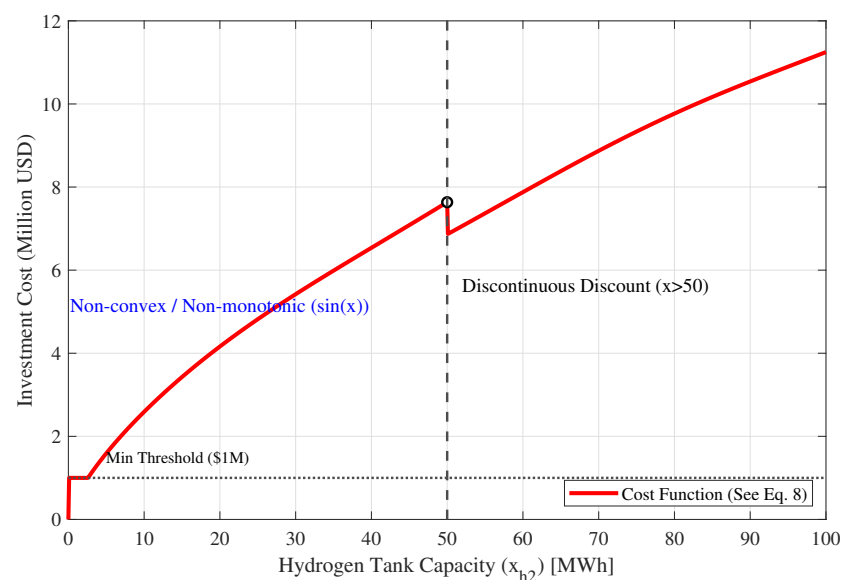
is applied when the capacity exceeds a critical threshold (50 MWh), introducing a discontinuity in the marginal cost:

$$C^{\text{base}}(E_{H2}) = \begin{cases} \kappa_{H2} \cdot (E_{H2})^{0.7}, & \text{if } 0 < E_{H2} \leq 50 \\ [\kappa_{H2} \cdot (E_{H2})^{0.7}] \cdot 0.9, & \text{if } E_{H2} > 50 \end{cases} \quad (10)$$

- (2) **Cyclic Modularity Cost (Highly Non-convex):** This component introduces a controlled, non-convex perturbation to the cost surface. It is not claimed to be a first-principles engineering cost law for hydrogen tanks. Instead, it is used as a stylized numerical proxy to represent irregular cost perturbations that may arise from land constraints, modular procurement, installation logistics, or supplier-specific pricing rules. The purpose of this term is to construct a reproducible black-box benchmark on which the proposed algorithm can be tested:

$$C^{\text{cyc}}(E_{H2}) = \kappa_{\text{cyc}} \cdot \sin\left(\frac{E_{H2}}{10}\right) \quad (11)$$

- (3) **Parameter Settings:** The coefficient expressed as  $\kappa_{H2} = 500 \times 10^3$  \$/MWh is selected to keep the tank cost on the same order of magnitude as the other investment terms. The exponent of 0.7 is used to emulate economies of scale, a standard sub-linear behavior in capacity–cost relationships, whereas the factor of 0.9 represents a stylized bulk-purchase discount beyond 50 MWh. The perturbation amplitude is set to  $\kappa_{\text{cyc}} = 100 \times 10^3$  \$, and the minimum entry cost is  $C_{\text{min}} = 1 \times 10^6$  \$. These values should be interpreted as benchmark settings for methodological evaluation rather than universally valid equipment prices.



**Figure 2.** Black-box function.

#### 2.4. Sub-Problem (SP): Multi-Scenario Operational Model

The Sub-Problem (SP) is formulated as an *explicit* Linear Programming (LP) problem. It is solved for each realized scenario ( $\xi_s$ ), given the fixed investment decisions ( $x$ ) determined in Stage I.



### 2.4.1. Operational Decision Variables ( $y_s$ )

For each scenario ( $s$ ) and time interval ( $t \in \mathcal{T} = \{1, \dots, 24\}$ ), the operational vector includes the following:

- $p_{s,t}^{\text{WT}}$ : Actual dispatched wind power (MWh);
- $p_{s,t}^{\text{EL}}$ : Power consumption of the electrolyzer (MWh);
- $m_{s,t}^{\text{H2}}$ : Hydrogen production rate ( $\text{Nm}^3$ );
- $S_{s,t}^{\text{H2}}$ : State of Charge (SoC) of the hydrogen tank ( $\text{Nm}^3$ );
- $p_{s,t}^{\text{grid,buy}} / p_{s,t}^{\text{grid,sell}}$ : Power purchased from/sold to the utility grid (MWh);
- $p_{s,t}^{\text{FC}}$ : Power generation from the fuel cell (MWh);
- $\sigma_{s,t}$ : Vector of slack variables for load shedding (ensuring feasibility).

### 2.4.2. Objective Function

The objective is to minimize the annualized operational cost for scenario  $s$ , calculated as the daily cost extrapolated to a full year (365 days):

$$Q(x, \xi_s) = 365 \cdot \sum_{t=1}^{24} \left( \lambda_t^{\text{buy}} p_{s,t}^{\text{grid,buy}} - \lambda_t^{\text{sell}} p_{s,t}^{\text{grid,sell}} + \rho^{\text{shed}} \sigma_{s,t}^{\text{load}} \right) \quad (12)$$

where  $\lambda_t$  represents the time-of-use electricity price and  $\rho^{\text{shed}}$  is the penalty coefficient for load shedding.

### 2.4.3. Constraints

The optimization is subject to the following technical constraints for  $\forall t \in \mathcal{T}$ :

- (1) Power Balance Constraint: The system must maintain real-time power equilibrium. The sum of generation and imports must equal the sum of consumption and exports:

$$p_{s,t}^{\text{WT}} + p_{s,t}^{\text{FC}} + p_{s,t}^{\text{grid,buy}} = p_{s,t}^{\text{EL}} + p_{s,t}^{\text{grid,sell}} + p_{s,t}^{\text{load}} - \sigma_{s,t}^{\text{load}} \quad (13)$$

- (2) Hydrogen Mass Balance Constraint: The hydrogen storage level evolves according to production, fuel-cell consumption, exogenous hydrogen demand, and storage loss.  $\eta^{\text{EL}}$  and  $\eta^{\text{FC}}$  denote the energy-to-volume and volume-to-electricity conversion coefficients, respectively.  $\lambda^{\text{loss}}$  is the hourly storage-loss coefficient:

$$m_{s,t}^{\text{prod}} = \eta^{\text{EL}} p_{s,t}^{\text{EL}}, \quad (14)$$

$$m_{s,t}^{\text{loss}} = \lambda^{\text{loss}} S_{s,t-1}^{\text{H2}}, \quad (15)$$

$$S_{s,t}^{\text{H2}} = S_{s,t-1}^{\text{H2}} + m_{s,t}^{\text{prod}} - \frac{p_{s,t}^{\text{FC}}}{\eta^{\text{FC}}} - D_{s,t}^{\text{H2}} - m_{s,t}^{\text{loss}}. \quad (16)$$

The storage level must also satisfy the cyclic boundary condition ( $S_{s,0}^{\text{H2}} = S_{s,24}^{\text{H2}}$ ). In the numerical base case,  $\lambda^{\text{loss}}$  is set to zero because reliable leakage data are not available; however, the  $m_{s,t}^{\text{loss}}$  variable is retained in the formulation so that boil-off, leakage, or purification losses can be activated directly when engineering data are available.

- (3) Investment Capacity Constraints: Operational variables are strictly bounded by the installed capacities determined in Stage I ( $x$ ):

$$0 \leq P_{s,t}^{\text{WT}} \leq \zeta_{s,t}^{\text{wind}} \cdot P_{\text{rated}}^{\text{WT}} \cdot x_{\text{WT}} \quad (17a)$$

$$0 \leq P_{s,t}^{\text{EL}} \leq P_{\text{EL}} \quad (17b)$$

$$0 \leq S_{s,t}^{\text{H}_2} \leq \eta^{\text{vol}} \cdot E_{\text{H}_2} \quad (17c)$$

$$0 \leq P_{s,t}^{\text{FC}} \leq P_{\text{FC}} \quad (17d)$$

Note: Equation (17c) includes  $\eta^{\text{vol}}$  to convert the decision variable ( $E_{\text{H}_2}$ , MWh) into storage volume limits ( $\text{Nm}^3$ ). All operational variables are non-negative ( $y_s \geq 0$ ).

## 2.5. Assumptions, Scope, and Limitations

The proposed case study is designed to evaluate the solution strategy rather than to deliver a full techno-economic or environmental assessment of a specific project. The main assumptions, their expected impacts, and possible extensions are summarized as follows.

- **System scope.** The studied WH-IES includes wind generation, electrolyzer capacity, hydrogen storage, and fuel-cell conversion. Photovoltaic generation and other storage technologies are not included. This simplification isolates the wind–hydrogen investment trade-off, but it may underestimate the value of multi-energy complementarity in sites with strong solar resources.
- **Wind representation.** The operational model uses scenario-based available wind power as exogenous input data. Detailed aerodynamic turbine modeling is omitted because the focus is planning under uncertainty and black-box investment costs, not turbine performance identification.
- **Hydrogen conversion and losses.** Electrolyzer and fuel-cell processes are represented by linear conversion coefficients in the second-stage LP. This preserves tractability across many scenarios and hourly periods, but it neglects part-load efficiency, start-up dynamics, temperature effects, and degradation. Hydrogen storage loss is represented by  $m_{s,t}^{\text{loss}}$  and  $\lambda^{\text{loss}}$  in the revised mass-balance equation; the base case sets  $\lambda^{\text{loss}} = 0$  due to the absence of site-specific leakage data.
- **Component lifetime and cycling.** The base case uses a 10-year economic recovery horizon for comparability with the original numerical experiment. This is shorter than the possible physical lifetime of wind assets, which can reach 20–30 years with proper maintenance and life-extension decisions [28]. Electrolyzer and fuel-cell replacement is not modeled explicitly. A practical extension is to introduce equivalent operating-hour or cycle constraints, for example,

$$\sum_{d=1}^{365} \sum_{t \in T} u_{d,t}^{\text{EL}} \Delta t \leq H_{\text{EL}}^{\text{life}}, \quad (18)$$

where  $u_{d,t}^{\text{EL}}$  is the electrolyzer operating indicator and  $H_{\text{EL}}^{\text{life}}$  is the allowable equivalent operating hours before stack replacement. Replacement costs can then be added to  $C_{\text{EL}}(P_{\text{EL}})$  through a component-specific replacement factor.

- **Financial scope.** Taxes, inflation, salvage value, and detailed replacement schedules are not included. The sensitivity in Section 4.6 explains how the annualization horizon affects economic interpretation.
- **Environmental scope.** The present manuscript focuses on economic planning and algorithmic tractability.  $\text{CO}_2$  accounting, life-cycle impacts, LCOE/LCOH<sub>2</sub>, and social–environmental externalities are not included in the objective function. A simple extension is to add an emission term or a carbon price to the operating cost, as discussed in Section 4.6.



- Methodological scope. The sinusoidal perturbation in the hydrogen-tank cost is a stylized black-box benchmark. It should be interpreted as a proxy for irregular non-analytical pricing, not as a universal engineering cost model.

### 3. Solution Methodology: The GSOA-Benders Framework

As previously described, the investment cost function introduced in the Master Problem (MP) constitutes an implicit, non-convex, and non-monotonic function characterized by sinusoidal terms. Such characteristics render the problem mathematically intractable for traditional “monolithic” MINLP solvers (e.g., Gurobi [17]).

To address this computational challenge, this paper proposes a novel hybrid heuristic–Benders decomposition framework, denoted as GSOA-Benders. The fundamental philosophy of this framework is decoupling:

1. Preservation of Explicit Sub-structures: The multi-scenario operational Sub-Problem (SP) remains an explicit, highly structured Linear Programming (LP) problem. Consequently, we retain the use of precise mathematical programming solvers (specifically, the dual-simplex algorithm) to ensure computational efficiency and optimality for this stage.
2. Isolation of Implicit Complexity: The implicit “black-box” investment decision process within the master problem is isolated from the linear definitions. We employ the General Soldiers Optimization Algorithm (GSOA)—a metaheuristic specifically engineered for such derivative-free landscapes—to solve the MP.

#### 3.1. Hybrid Heuristic Decomposition Framework

The iterative solution procedure of the proposed GSOA-Benders framework is illustrated in Figure 3. The core mechanism is theoretically grounded as follows:

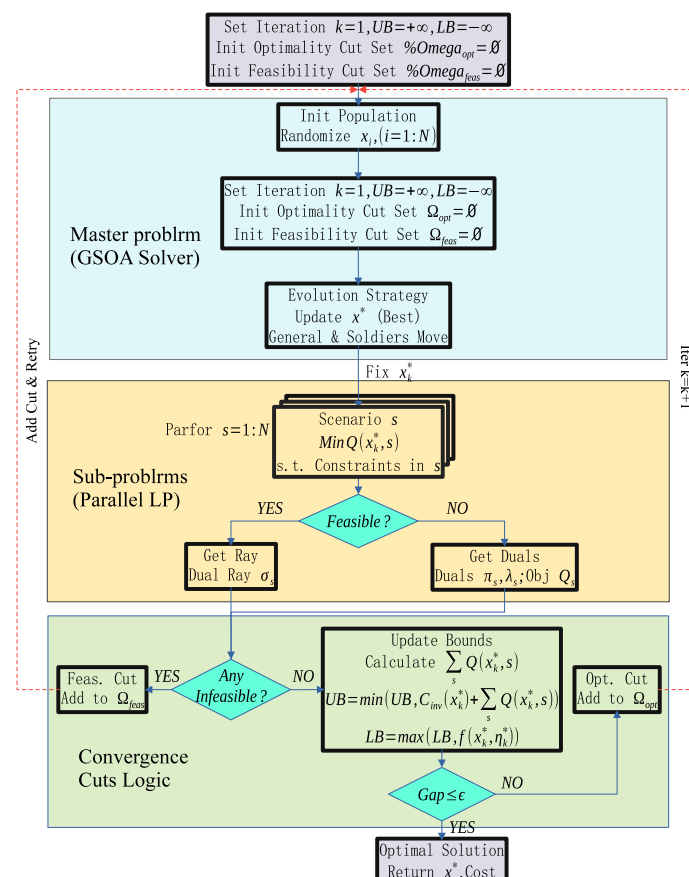


Figure 3. Overall flowchart of the GSOA-Benders framework.

Classically, Benders decomposition is applied to problems in which the master problem can be solved exactly or to proven optimality within a convex or mixed-integer framework. The present problem is different because the first-stage term ( $C_{inv}(\mathbf{x})$ ) may be non-convex and black-box in nature. Therefore, the theoretical role of Benders cuts must be stated carefully.

- Recourse convexity. For a fixed scenario ( $s$ ), the operational subproblem can be written in the generic LP form:

$$Q_s(\mathbf{x}) = \min_{\mathbf{y}_s} \left\{ \mathbf{c}_s^\top \mathbf{y}_s \mid \mathbf{W}_s \mathbf{y}_s \geq \mathbf{h}_s - \mathbf{T}_s \mathbf{x}, \mathbf{y}_s \geq 0 \right\}. \quad (19)$$

Under complete recourse or with sufficiently penalized slack variables, the subproblem is feasible for all  $\mathbf{x} \in \Omega$ . According to LP duality,  $Q_s(\mathbf{x})$  is the pointwise maximum of affine functions in  $\mathbf{x}$  and is therefore convex and piecewise linear. Consequently, the expected recourse function, i.e.,

$$\bar{Q}(\mathbf{x}) = \sum_{s \in \mathcal{S}} \pi_s Q_s(\mathbf{x}), \quad (20)$$

is also convex and admits supporting hyperplanes generated from optimal dual solutions.

- Benders cut validity. Let  $\lambda_s^{(k)}$  be an optimal dual vector of scenario  $s$  at the incumbent investment vector ( $\mathbf{x}^{(k)}$ ). A valid expected recourse cut is

$$\vartheta \geq \sum_{s \in \mathcal{S}} \pi_s \left[ Q_s(\mathbf{x}^{(k)}) + \left( \mathbf{g}_s^{(k)} \right)^\top (\mathbf{x} - \mathbf{x}^{(k)}) \right], \quad \mathbf{g}_s^{(k)} = -\mathbf{T}_s^\top \lambda_s^{(k)}. \quad (21)$$

This cut lower-bounds only the expected second-stage operating cost ( $\bar{Q}(\mathbf{x})$ ). It does not convexify the black-box investment function ( $C_{inv}(\mathbf{x})$ ) and therefore does not, by itself, prove the global optimality of the total objective.

- Stability-gap definition. Because GSOA is a derivative-free metaheuristic, the master problem is not solved with a deterministic global optimality guarantee. We therefore report a stability gap rather than a strict optimality gap. At iteration  $k$ , the best feasible upper bound is

$$UB_k = \min_{i \leq k} \left\{ C_{inv}(\mathbf{x}^{(i)}) + \bar{Q}(\mathbf{x}^{(i)}) \right\}. \quad (22)$$

Let  $\widehat{LB}_k$  be the best master value found by the GSOA over the current cut approximation. The stability gap is defined as

$$G_k^{stab} = \frac{|UB_k - \widehat{LB}_k|}{\max\{1, |UB_k|\}}. \quad (23)$$

The stopping rule is  $G_k^{stab} \leq \varepsilon$ , together with no material improvement in  $UB_k$  for a fixed number of iterations. This criterion indicates that the incumbent solution is stable with respect to the accumulated cut-plane approximation, but it should not be interpreted as a mathematical certificate of global optimality for a non-convex black-box master problem.

- Solution-quality safeguards. To improve reliability, the implementation uses bounded decision domains, feasibility cuts when subproblem infeasibility is detected, repeated GSOA starts, and ex-post evaluation of the final investment vector by solving all scenario subproblems. These safeguards make the method suitable as an engineering decision-support procedure, while the limitation regarding global optimality is explicitly acknowledged.

### 3.2. Master Problem Solver: The GSOA Algorithm

The pivotal innovation of the GSOA-Benders framework lies in the strategic selection of the solver for the Master Problem (MP).

Diverging from the conventional reliance on commercial solvers such as Gurobi, this study adopts the novel General Soldiers Optimization Algorithm (GSOA). The GSOA is a population-based metaheuristic designed to operate as a generic “black-box” solver, offering distinct advantages in this context. The full pseudocode of the proposed GSOA–Benders framework is provided in Appendix A.

Instead of relying solely on commercial solvers such as Gurobi, this study employs the General Soldiers Optimization Algorithm (GSOA) to solve the black-box master problem. As a population-based metaheuristic, GSOA can evaluate candidate investment decisions without requiring an explicit analytical reformulation of the cost function. The full pseudocode of GSOA is provided in Appendix A, as suggested by the reviewers, while the main text summarizes only its role in the proposed GSOA–Benders framework.

1. **Model Agnosticism:** The GSOA does not mandate the objective function to be differentiable, continuous, or convex. It requires solely a “Fitness Function” mechanism that accepts an input decision vector ( $\mathbf{x}$ ) and evaluates it to return a scalar cost value.
2. **Superior Flexibility:** This characteristic enables the fitness function to directly encapsulate the “black-box” logic—specifically, the sinusoidal terms ( $\sin(x)$ ) detailed in Equation (11). This represents a modeling capability that is fundamentally unattainable by standard gradient-based solvers.
3. **Mixed-Integer Handling Capability:** The GSOA seamlessly processes both binary variables (e.g.,  $x_{WT} \in \{0, 1\}$ ) and continuous variables (e.g.,  $P_{EL}$ ) simultaneously, utilizing simple boundary constraints and discrete rounding mechanisms.

### 3.3. Sub-Problem Solution and Expected Cut Generation

In Step 2 of each Benders iteration, the framework evaluates the expected operational cost of the candidate investment solution ( $\mathbf{x}^{(k)}$ ) across all  $N$  uncertain scenarios.

#### 3.3.1. Sub-Problem (SP) Solution

The sub-problem maintains a standard LP structure. We iterate through all scenarios ( $s \in \mathcal{S}$ ), passing the fixed investment variables ( $\mathbf{x}^{(k)}$ ) and the scenario-specific parameters ( $\xi_s$ ) to the LP solver.

#### 3.3.2. Expected Cut Generation

Upon solving  $N$  sub-problems, we obtain  $N$  sets of operational costs ( $Q(\mathbf{x}^{(k)}, \xi_s)$ ) and their corresponding dual-variable vectors ( $\lambda_s$ ). We construct a single expected Benders cut via probability-weighted averaging to provide feedback to the master problem:

- **Expected Operational Cost:**

$$\bar{Q}^{(k)} = \sum_{s=1}^N \pi_s \cdot Q(\mathbf{x}^{(k)}, \xi_s) \quad (24)$$

- **Expected Dual Variables:**

$$\bar{\lambda}^{(k)} = \sum_{s=1}^N \pi_s \cdot \lambda_s \quad (25)$$

This “Average Cut” encapsulates the expected performance of the investment plan ( $\mathbf{x}^{(k)}$ ) under all uncertainties, thereby guiding the GSOA to search towards statistically stable decision regions.

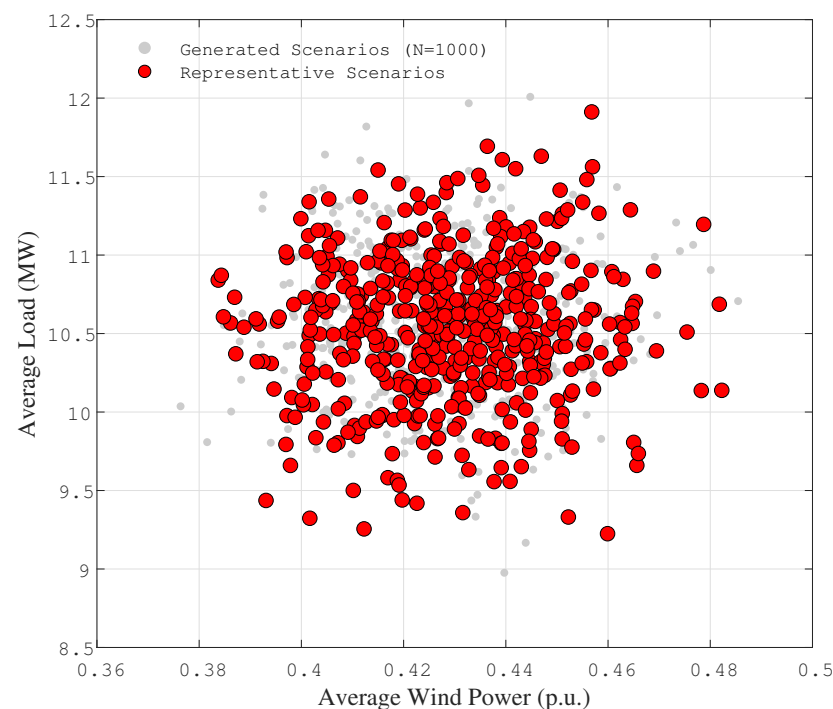
## 4. Case Study

To comprehensively validate the effectiveness, computational efficiency, and robustness of the proposed GSOA-Benders framework in handling intractable black-box cost functions, this section presents two key comparative experiments.

### 4.1. Experimental Setup

#### 4.1.1. Uncertainty Scenario Generation

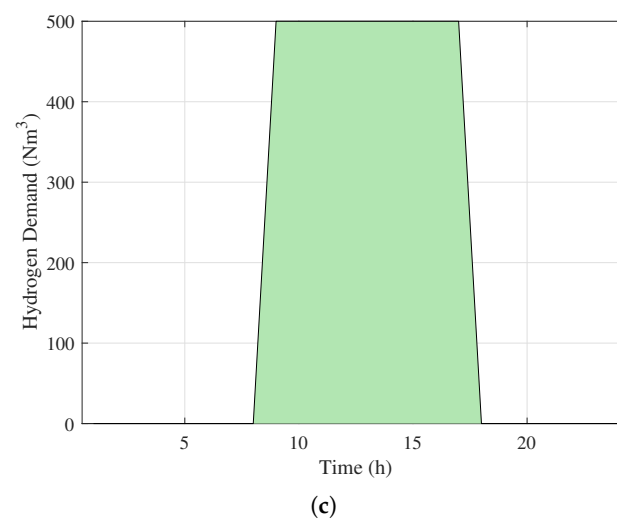
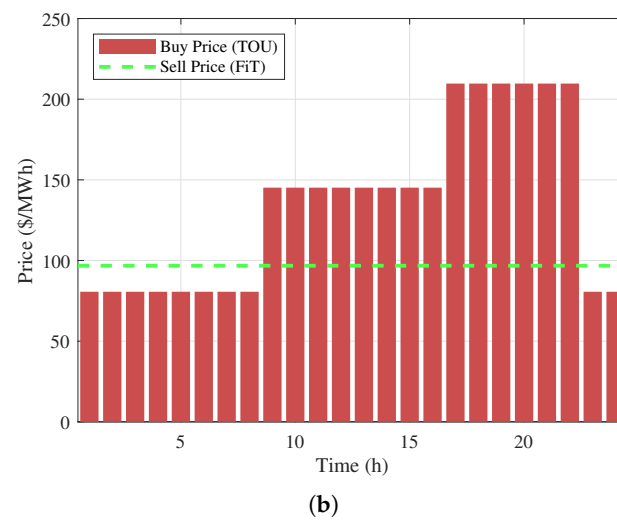
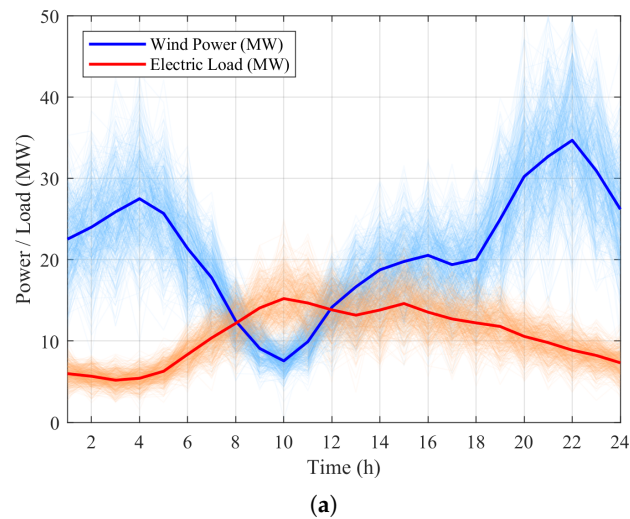
To effectively capture the stochastic nature of wind power generation and electrical load demand, we employed Monte Carlo simulation to generate an initial set of  $N_{\text{initial}} = 1000$  stochastic operational scenarios [29]. Subsequently, the scenario-reduction procedure illustrated in Figure 4, was applied to extract the  $N = 500$  most representative scenarios for the optimization.



**Figure 4.** Scenario distribution and reduction.

As illustrated in Figure 4, the original scenario set (depicted as gray scatter points) covers a broad wind-load distribution space, while the red points represent the selected typical scenarios. This visualization validates the representativeness of the scenario generation process.

Figure 5a illustrates the temporal distribution of these 500 stochastic scenarios, where the red and blue envelopes depict the fluctuation ranges for electrical load and wind power generation, respectively. Figure 5b presents the comparative profiles of the Time-of-Use (TOU) electricity tariff versus the Feed-in Tariff (FiT). Figure 5c specifies the inflexible hydrogen demand profile, which must be strictly satisfied during the daily operational window from 09:00 to 17:00.



**Figure 5.** Input data for the case study: (a) uncertainty scenarios; (b) grid price; (c) H<sub>2</sub> demand.

#### 4.1.2. Hardware Platform

All numerical experiments were conducted on a unified computing platform to ensure fair comparison. The hardware configuration consists of a 12th Gen Intel(R) Core(TM) i5-12490F processor (12 logical processors) and 16GB RAM, running MATLAB R2023b and Gurobi 12.0.3.

#### 4.1.3. Benchmark Algorithms

To provide a transparent evaluation, the proposed framework is benchmarked against several representative alternatives. These benchmarks are not claimed to exhaust the current state of the art; instead, they are selected to contrast explicit-solver, reformulation-based, exhaustive-heuristic, and heuristic-decomposition paradigms:

- **Algorithm A (Benchmark 1): Gurobi–Monolithic**  
The traditional “monolithic” approach consolidates the master problem and all  $N$  operational sub-problems into a single, integrated Mixed-Integer Non-Linear Programming (MINLP) model and attempts to solve it directly using the Gurobi 12.0 solver.
- **Algorithm B (Benchmark 2): MILP-Benders (Reformulation) [20]**  
The “mathematical reformulation” approach attempts to approximate the black-box costs (e.g., step-wise costs) by reformulating them into explicit Mixed-Integer Linear Programming (MILP) constraints (e.g., via Big-M techniques) that are solvable by Gurobi.
- **Algorithm C (Benchmark 3): GSOA + Simulation (Exhaustive)**  
The “simulation-based optimization” approach, unlike the proposed framework, does not utilize Benders cuts. Instead, for every fitness evaluation in the master problem, the GSOA performs an exhaustive loop over all  $N$  scenarios by calling the subproblem routine to compute the actual operational cost.
- **Algorithm D (Benchmark 4): SFOA-Benders [30]**  
As a comparative heuristic decomposition framework, this approach shares an identical structural architecture with GSOA-Benders but substitutes the master problem solver with the Starfish optimization algorithm (SFOA) [31] to evaluate the impact of the specific heuristic engine.
- **Algorithm E (Proposed): GSOA-Benders**  
This approach comprises the novel hybrid heuristic decomposition framework established in Section 3.

#### 4.2. Experiment I: Explicit Non-Convex Problem ( $N = 1000$ )

First, it is imperative to conduct an unbiased evaluation of all algorithms on an *explicit*, non-convex problem that is mathematically modelable by commercial solvers.

For this baseline experiment, the scenario count is set to  $N = 1000$ . The investment cost function is configured as a standard analytical form: wind power involves binary decisions ( $x_{WT} \in \{0, 1\}$ ), while the hydrogen-tank cost follows an explicit, non-linear power function ( $C \propto x^{0.6}$ ). The objective of this setup is to probe the performance limits of Gurobi and to benchmark the proposed GSOA-Benders framework against a solvable explicit baseline.

As evidenced in Table 1, the Gurobi 12.0 solver demonstrated superior performance. Driven by its advanced presolve capabilities (eliminating one-quarter of the problem scale within 0.43 s) and its highly efficient MINLP branch-and-bound logic, Gurobi decisively outperformed all benchmark algorithms, completing the optimization task in just 36.58 s.

**Table 1.** Comparison of algorithm performance.

| Algorithm    | Variables/Constraints | Time (s) | Optimal Cost |
|--------------|-----------------------|----------|--------------|
| Gurobi       | 240,005/192,000       | 36.58    | −381,058.9   |
| GSOA-Benders | 4/240                 | 45.66    | −342,939.52  |

This result corroborates a fundamental premise: for any *explicit* mathematical problem capable of being formulated within the syntax of commercial solvers, the monolithic



solver remains the optimal choice. It is acknowledged that the proposed GSOA-Benders framework exhibits no advantage in terms of either computational speed or solution precision for such analytically tractable problems.

#### 4.3. Experiment II: The “Intractable” Black-Box Problem ( $N = 500$ )

This case study focuses on the “Intractable Black-Box” problem. The scenario count is set to  $N = 500$ . The investment cost configuration is modified as follows: wind power retains binary decisions, while the hydrogen-tank cost is replaced by the “Intractable Black-Box” function (incorporating  $x^{0.7}$  and  $\sin(x)$  terms). The calculation results are shown in Table 2.

**Table 2.** Benchmark comparison of different algorithms.

| Algorithm | Solution Status                | Time (s) | Cost        |
|-----------|--------------------------------|----------|-------------|
| A         | Direct exact solve unavailable | 0.00     | (N/A)       |
| B         | Direct exact solve unavailable | 0.00     | (N/A)       |
| C         | Failed (Timeout)               | >17,500  | (N/A)       |
| D         | Successfully Converged         | 51.15    | −242,940.18 |
| E         | Successfully Converged         | 35.86    | −242,940.18 |

##### 4.3.1. Modeling Infeasibility of Gurobi and MILP Benchmarks

Both Gurobi–Monolithic and MILP-Benders encountered obstacles at the initial modeling stage when applied directly to the non-analytical benchmark.

It is acknowledged that, theoretically, techniques such as piecewise linearization (e.g., using SOS2 constraints) could be employed to approximate non-convex functions (like  $x^{0.7}$ ), thereby rendering the problem solvable by Gurobi. However, in this specific context, such an approach is practically infeasible due to the following reasons:

- **Intractability of Approximation:** To approximate the high-frequency sinusoidal component with sufficient precision, a piecewise linear formulation would require an excessive number of binary variables.
- **Fundamental Infeasibility:** More critically, if the cost function were a true black box (e.g., a compiled external simulation or a neural network), explicit reformulation would be mathematically impossible, rendering direct application of a conventional explicit solver impossible unless an additional surrogate or approximation layer is introduced.

In summary, faced with the coupling of high-frequency fluctuations and large-scale scenarios, attempting to transform the “Intractable Black Box” into an “explicit” model via mathematical reformulation is computationally unrealistic.

##### 4.3.2. Computational Intractability of Simulation-Based Optimization

While the GSOA + Simulation framework can successfully model “black-box” costs, its computational burden is prohibitive. A standard GSOA run (e.g., 100 iterations, 50 particles) requires 5000 fitness evaluations. With  $N = 500$  scenarios, each evaluation necessitates the solution of 500 Linear Programs (LPs). The estimated computation time is derived as follows:

$$\begin{aligned}
 T_{total} &\approx 5000 \text{ evals} \times (500 \text{ LPs} \times 0.007 \text{ s/LP}) \\
 &\approx 17,500 \text{ s} \approx 4.86 \text{ hours}
 \end{aligned}
 \tag{26}$$

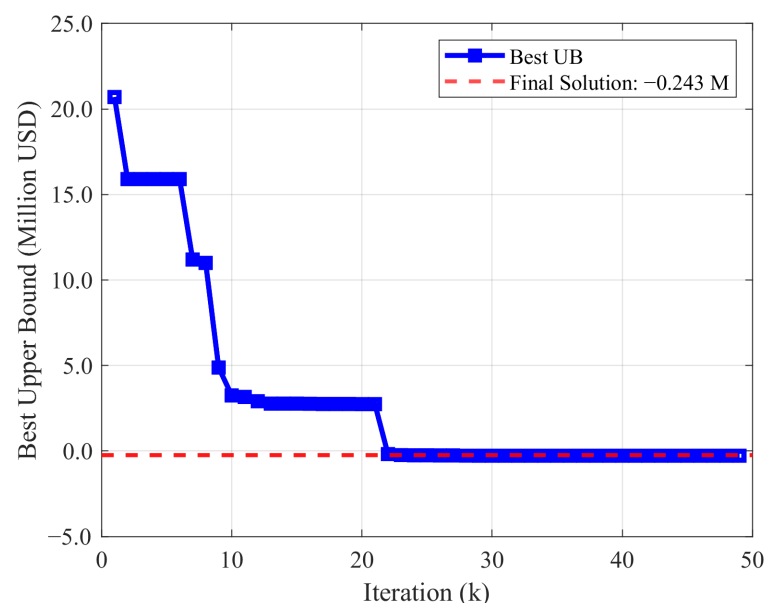
Consequently, while the “Simulation + Optimization” framework offers flexibility, it is computationally prohibitive for multi-scenario stochastic problems.

#### 4.3.3. Performance Superiority of the Proposed Framework

Among the evaluated methods, GSOA-Benders provides a viable framework possessing both “flexibility” and “efficiency” for the tested black-box setting.

- Flexibility: Our black-box solver successfully handled the sinusoidal costs ( $\sin(x)$ ) that Gurobi failed to model natively.
- Efficiency: By utilizing Benders cuts as a dynamic “surrogate model”, our framework completed the optimization task in 35.86 s. This represents a substantial speedup over the estimated 4.86 h required by the “Simulation + Optimization” approach under the same LP-call assumptions. Furthermore, compared to the SFOA-Benders benchmark, the proposed GSOA-Benders achieved a  $1.43\times$  speedup.

As illustrated by the convergence curve in Figure 6, the proposed GSOA-Benders framework successfully converged within 36 s. It provides a practical tool for solving “implicit”, multi-scenario planning problems when direct exact formulation is not available. This supports the role of the proposed framework as a bridge between difficult-to-formulate mathematical programming models and computationally expensive pure heuristics while not implying universal superiority over exact solvers for explicit models.



**Figure 6.** Convergence curve.

#### 4.4. Economic Analysis and Insights

The optimal investment solution converged upon by the GSOA-Benders framework is denoted as

$$\mathbf{x}^* = [1, 0.53, 23.23, 0]^\top, \quad (27)$$

corresponding to the capacity of the Wind Turbine ( $x_{WT}$ , binary), Electrolyzer ( $P_{EL}$ , MW), Hydrogen Tank ( $E_{H2}$ , MWh), and Fuel Cell ( $P_{FC}$ , MW), respectively. This result reveals profound economic insights driven by the solver’s interaction with the system physics and cost structures.

##### 4.4.1. Wind Power Dominance vs. Fuel-Cell Viability

1. Wind Power ( $x_{WT} = 1$ ): The algorithm identified the construction of the 50 MW wind farm as the primary revenue driver, prioritizing electricity sales to the grid to maximize profit.

2. Fuel Cell ( $P_{FC} = 0$ ): Conversely, the solution confirms that the “Power-to-Gas-to-Power” (P2G2P) round-trip energy storage arbitrage is economically unviable under the current cost parameters. The efficiency losses in the hydrogen–electricity conversion loop outweigh the arbitrage revenue. Consequently, the algorithm strictly avoided any capital expenditure on fuel cells.

#### 4.4.2. Strategic Exploitation of Non-Convex Cost Structures ( $E_{H2} = 23.23$ )

The decision to invest in a specific capacity of 23.23 MWh for the hydrogen tank represents a sophisticated decision-making outcome. The GSOA solver successfully navigated the non-convex cost landscape to identify a precise trade-off:

1. Exploiting Local Cost Minima: Unlike simple step-wise costs, the intractable black-box function (incorporating  $\sin(x)$  terms) creates specific local minima or “cost valleys”. The algorithm detected that the capacity of 23.23 MWh coincides with a valley in the sinusoidal cost curve, offering a distinct marginal cost advantage.
2. Balancing Scale and Operational Flexibility: The algorithm determined that at this specific capacity, the benefits of economies of scale (driven by the  $x^{0.7}$  power term), combined with the provided scenario-based operational flexibility (smoothing fluctuations across 500 scenarios) outweighed the marginal investment cost.
3. Discovery of Counterintuitive Solutions: This validates a critical capability, i.e., that the GSOA-Benders framework does not merely “accommodate” the complexity of black-box functions; it actively exploits their non-convex characteristics to discover superior, counterintuitive solutions that traditional linear approximations might overlook.

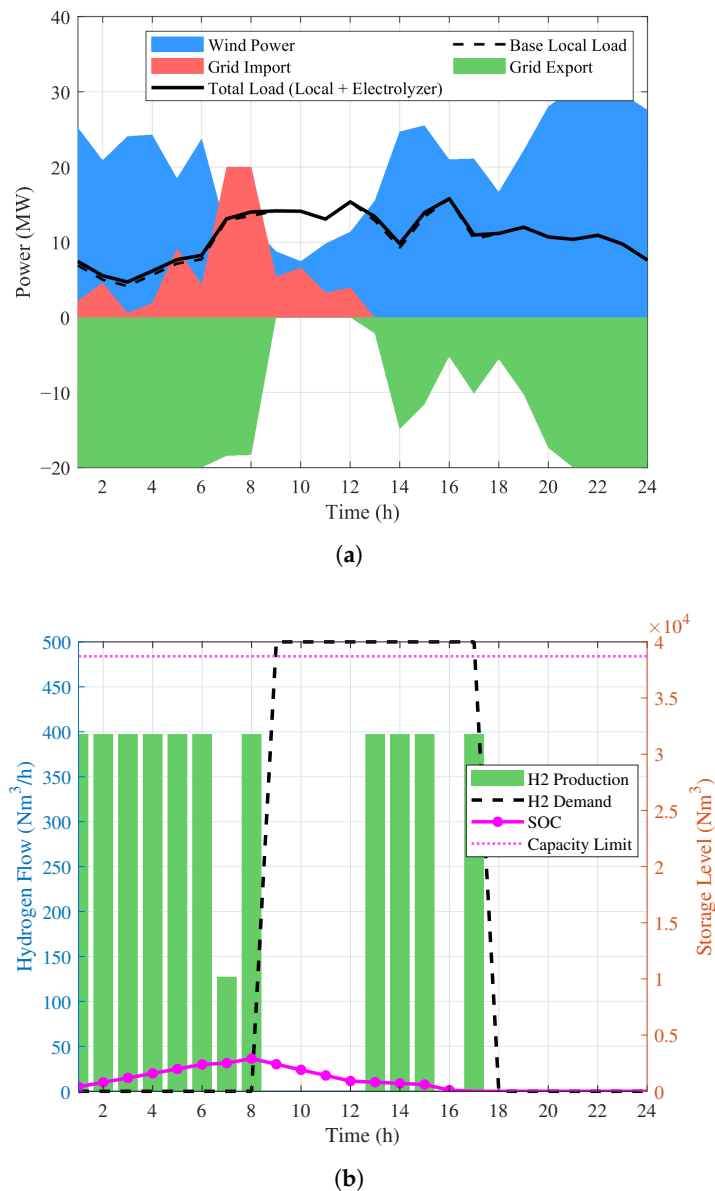
#### 4.5. Operational Dispatch Analysis

Figure 7 illustrates the optimal dispatch strategy under a representative stochastic scenario.

##### 4.5.1. Electricity-Side Operation Analysis

Figure 7a depicts the power supply–demand balance, revealing the complex interaction mechanism between wind generation, hydrogen production, and grid interaction:

1. Wind Accommodation and Revenue Generation: The blue region represents wind power generation. During most periods (especially at night and early morning), wind output significantly exceeds the system’s internal load (solid black line). This surplus power is exported to the grid (green region below the X-axis), generating substantial Feed-in Tariff (FiT) revenue. This validates the algorithm’s strategic preference for maximizing wind power investment in Stage I.
2. Price-Responsive Electrolyzer Scheduling: Observing the deviation between the total load (solid black line) and the base load (dashed black line), it is evident that electrolyzer operation is concentrated in the 00:00–08:00 and 22:00–24:00 intervals. This strategy is economically driven. The algorithm identifies these windows as having both abundant wind resources and off-peak electricity prices, making them the most cost-effective periods for hydrogen production.
3. Strategic Grid Imports (Pre-charging): Notably, during intervals such as 07:00–09:00, significant grid power purchases (red region) occur despite the presence of wind generation. This is because the electrolyzer maintains high-power operation to accumulate sufficient hydrogen reserves before the onset of high-price periods. Consequently, the total load temporarily exceeds wind output. This strategy of “purchasing power to store hydrogen” (strategic pre-charging) while increasing instantaneous costs avoids the necessity of producing hydrogen during peak-price hours or incurring penalty costs for shortages later in the day, thereby achieving global optimality over the daily horizon.



**Figure 7.** Optimal operation dispatch: (a) power balance showing wind integration; (b) hydrogen balance showing the smoothing effect of the storage tank.

#### 4.5.2. Hydrogen-Side Operation Analysis

Figure 7b elucidates the “peak shaving and valley filling” mechanism of the hydrogen system:

1. **Temporal Supply–Demand Mismatch:** Hydrogen production (green bars) is concentrated at night, whereas the hydrogen demand (black dashed line) is rigidly distributed during daytime working hours (09:00–17:00). This creates a significant temporal mismatch between supply and demand.
2. **Buffering Role of Storage Tank:** The pink line represents the State of Charge (SoC) of the hydrogen tank, illustrating four distinct phases:
  - **Charging Phase (00:00–08:00):** As the electrolyzer runs at full capacity, the SoC rises continuously, converting surplus wind energy into stored hydrogen.
  - **Standby Phase (08:00–09:00):** Hydrogen production ceases. The SoC remains high, effectively “standing by” for the upcoming peak demand.

- Discharging Phase (09:00–17:00): During the rigid demand window, the electrolyzer shuts down to avoid high electricity prices. The system relies entirely on the storage tank to satisfy demand. Consequently, the SoC curve exhibits a near-linear depletion trajectory.
  - Depletion Phase (17:00–18:00): By the time demand ends, the SoC drops to near-zero levels.
3. High-Capacity Utilization: It is noteworthy that the peak SoC reaches approximately  $4 \times 10^3 \text{ Nm}^3$  (corresponding to roughly 23 MWh), which aligns precisely with the investment decision derived in Stage I. This confirms that the algorithm not only calculated the optimal capacity but also fully utilized this capacity in the actual operational dispatch, leaving no redundant investment.

#### 4.6. Sensitivity, Scalability Boundary, and Environmental Extension

The reviewers correctly noted that several assumptions influence the economic interpretation of the results. Therefore, three points are clarified.

##### 4.6.1. Project Lifetime Sensitivity

The base case adopts  $L = 10$  years as a conservative economic recovery horizon. This does not imply that wind farms physically last only ten years. To show the direction of the effect, Table 3 reports annualization factors under the straight-line factor ( $1/L$ ) and, for comparison, the CRF with  $r = 8\%$ . A longer project horizon lowers the annualized capital burden and would generally make capital-intensive assets, such as wind turbines and storage, more attractive. Because a full re-optimization under each lifetime assumption requires the rerunning of the black-box planning algorithm, the present paper reports this as a sensitivity interpretation rather than claiming a new optimal investment vector for each lifetime.

**Table 3.** Annualization-factor sensitivity for different economic lifetimes.

| Parameter/Lifetime             | 10 Years              | 20 Years      | 30 Years            |
|--------------------------------|-----------------------|---------------|---------------------|
| Straight-line factor ( $1/L$ ) | 0.100                 | 0.050         | 0.033               |
| CRF ( $r = 8\%$ )              | 0.149                 | 0.102         | 0.089               |
| Interpretation                 | Conservative recovery | Typical asset | Wind-life extension |

##### 4.6.2. Scalability Boundary

The numerical experiments in this paper cover 500–1000 scenarios. Therefore, the empirical scalability claim is limited to this tested range. The potential to scale further comes from the separability of the LP subproblems: for each Benders iteration, the dominant computation is approximately proportional to  $N$  independent scenario LPs, which can be parallelized across processors or compute nodes. However, this manuscript does not claim that million-scenario performance has been empirically demonstrated. Larger networks, more devices, and larger scenario sets are important future tests.

##### 4.6.3. Emission-Reduction Extension

Although the current objective is economic, the same formulation can incorporate emission reduction. A simple extension is to add a carbon-cost term to the operating objective:

$$Q^{CO_2}(x, \xi_s) = Q(x, \xi_s) + 365 \sum_{t \in T} \rho^{CO_2} \left( e_t^{grid} p_{s,t}^{grid,buy} - e_t^{offset} p_{s,t}^{grid,sell} \right), \quad (28)$$

where  $\rho^{CO_2}$  is the carbon price,  $e_t^{grid}$  is the grid-emission factor, and  $e_t^{offset}$  is the credited emission factor of exported renewable electricity or substituted fossil-based hydrogen.

This term is linear in the operational variables and can be added without changing the Benders-cut structure. Detailed life-cycle CO<sub>2</sub> accounting and green hydrogen certification are left for future work.

## 5. Conclusions

This paper addresses the two-stage stochastic investment planning problem for Wind–Hydrogen Integrated Energy Systems (WH-IESs) under scenario-based uncertainty by proposing a hybrid heuristic–Benders framework, GSOA-Benders. By combining the derivative-free search capability of the General Soldiers Optimization Algorithm (GSOA) with Benders cuts generated from linear recourse subproblems, the proposed method provides a practical way to handle non-analytical first-stage investment costs while retaining efficient scenario decomposition.

The core conclusions drawn from this study are summarized as follows:

1. **Structural Limitations of Monolithic Solvers:** While commercial solvers like Gurobi demonstrate exceptional speed (solving  $N = 1000$  explicit scenarios in 36 s) for mathematically well-posed MINLP problems, they exhibit formulation rigidity. They are not intended to directly process “implicit”, “non-analytical”, and externally evaluated black-box cost functions unless an additional reformulation, approximation, or surrogate layer is introduced.
2. **Flexibility and Decomposition Efficiency of GSOA-Benders:** By decoupling the implicit black-box master problem from the explicit scenario subproblems, the proposed framework handles a class of problems where direct gradient-based or exact algebraic modeling is difficult. It serves as a practical bridge between complex engineering cost evaluators and structured mathematical optimization.
3. **Practical Viability:** Case studies show that for the tested problem involving  $N = 500$  scenarios and non-convex black-box cost functions, GSOA-Benders is a viable decision-support tool. It successfully converged within 36 s, identifying a high-quality stochastic investment strategy (yielding a total annualized cost of \$243k), thereby effectively balancing economic efficiency with operational reliability.

Looking forward, the proposed framework demonstrates potential in terms of extensibility and adaptability. Promising directions for future work include: (1) integration with AI-driven or simulation-driven cost estimators; (2) explicit modeling of component degradation, replacement, and life-cycle emissions; and (3) empirical scalability validation on larger networks and much larger scenario sets using high-performance parallel computing.

**Author Contributions:** Conceptualization, H.X., B.F. and Q.T.; methodology, H.X., B.F., Q.L. and Q.T.; formal analysis, H.X. and B.F.; investigation, H.X., F.Y., Y.K. and Y.H.; writing—original draft preparation, H.X. and Q.L.; writing—review and editing, H.X., B.F., F.Y., Y.K., Y.H., Q.L. and Q.T.; supervision, Q.T. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research was funded by the Science and Technology Project of State Grid Hubei Electric Power Co., Ltd., grant number B31532259621.

**Data Availability Statement:** The data presented in this study are available in the article. Additional data supporting the reported results are available from the corresponding author upon reasonable request.

**Acknowledgments:** The authors would like to thank the State Grid Hubei Electric Power Research Institute and Northwest A&F University for their support.

**Conflicts of Interest:** Authors Haozhe Xiong, Bingyang Feng, Fangbin Yan, Yiqun Kang and Yuxuan Hu were employed by the State Grid Hubei Electric Power Research Institute, State Grid Hubei Electric Power Co., Ltd. The remaining authors declare that the research was conducted in the absence of any

commercial or financial relationships that could be construed as a potential conflict of interest. The authors declare that this study received funding from the Science and Technology Project of State Grid Hubei Electric Power Co., Ltd. The funder was not involved in study design; the collection, analysis, or interpretation of data; the writing of this article; or the decision to submit for publication.

## Appendix A. GSOA Pseudocode

For completeness, the detailed General Soldiers Optimization Algorithm used as the master-problem search engine is provided below (Algorithm A1).

---

### Algorithm A1 General Soldiers Optimization Algorithm Used in the Master Problem

---

**Input:** Population size  $N_{pop}$ , Max iterations  $K_{max}$ , Bounds **LB**, **UB**, Objective function  $f(\cdot)$   
**Output:** Best solution  $\mathbf{x}^*$  and minimum cost  $Z^*$   
Initialize population  $\mathbf{X}$  randomly within bounds [**LB**, **UB**]  
Compute initial fitness  $F_i = f(\mathbf{x}_i)$  for all  $i$   
Set global best  $\mathbf{x}^*$  based on best fitness  
 $k \leftarrow 1$   
**while**  $k \leq K_{max}$  **do**  
    Sort population by fitness  $F$  (ascending order)  
    Identify Leaders:  $\mathbf{x}_{gen} \leftarrow \mathbf{x}_{rank(1)}$ ,  $\mathbf{x}_{L1} \leftarrow \mathbf{x}_{rank(2)}$ ,  $\mathbf{x}_{L2} \leftarrow \mathbf{x}_{rank(3)}$   
    Calculate mean position:  $\mathbf{x}_{mean} \leftarrow \frac{1}{N_{pop}} \sum_{i=1}^{N_{pop}} \mathbf{x}_i$   
    Update probability:  $GP \leftarrow 0.9 - 0.8(k/K_{max})$   
    **for**  $i = 1$  to  $N_{pop}$  **do**  
        Generate random numbers  $r_1, r_2, r_3 \in [0, 1]$   
        **if**  $r_1 < GP$  **then**  
            // Exploration Phase  
            **if**  $r_2 < 0.25$  **then**  
                 $\mathbf{x}_i^{new} \leftarrow \mathbf{x}_i + (\mathbf{x}_{gen} - \mathbf{x}_i) \cos(\theta)$   
            **else if**  $r_2 < 0.50$  **then**  
                 $\mathbf{x}_i^{new} \leftarrow \mathbf{x}_{mean} + \mathbf{d} \cdot w$   
            **else if**  $r_2 < 0.75$  **then**  
                 $\mathbf{x}_i^{new} \leftarrow \mathbf{x}_i + \mathcal{N}(0, 1)|\mathbf{x}_{L1} - \mathbf{x}_i|$   
            **else**  
                 $\mathbf{x}_i^{new} \leftarrow U \cdot \mathbf{x}_i + (1 - U)(\mathbf{x}_{avg\_rand} - \mathbf{x}_{mean})$   
            **end if**  
        **else**  
            // Exploitation Phase  
            Select 5 reference solutions to compute differential vector  $\mathbf{dm}$   
            **if**  $r_3 < GP$  **then**  
                 $\mathbf{x}_i^{new} \leftarrow \mathbf{x}_i + c_1 \mathbf{dm}_1 + c_2 \mathbf{dm}_2$   
            **else**  
                 $\mathbf{x}_i^{new} \leftarrow \mathbf{x}_i + (U \mathbf{dm}_1 + (1 - U)(\mathbf{dm}_2 - \mathbf{dm}_3))$   
            **end if**  
        **end if**  
        Boundary check:  $\mathbf{x}_i^{new} \leftarrow \min(\max(\mathbf{x}_i^{new}, \mathbf{LB}), \mathbf{UB})$   
        Compute fitness:  $F_{new} = f(\mathbf{x}_i^{new})$   
        **if**  $F_{new} < F_i$  **then**  
             $\mathbf{x}_i \leftarrow \mathbf{x}_i^{new}$ ;  $F_i \leftarrow F_{new}$   
            **if**  $F_{new} < Z^*$  **then**  
                 $\mathbf{x}^* \leftarrow \mathbf{x}_i^{new}$ ;  $Z^* \leftarrow F_{new}$   
            **end if**  
        **end if**  
    **end for**  
     $k \leftarrow k + 1$   
**end while**  
**return**  $\mathbf{x}^*, Z^*$

---



## References

- Gielen, D.; Boshell, F.; Saygin, D.; Bazilian, M.D.; Wagner, N.; Gorini, R. The role of renewable energy in the global energy transformation. *Energy Strategy Rev.* **2019**, *24*, 38–50.
- IRENA. *Renewable Energy Statistics 2025*; International Renewable Energy Agency: Abu Dhabi, The United Arab Emirates, 2025.
- IEA. *Electrolysers*; International Energy Agency: Paris, France, 2024.
- Kroposki, B.; Johnson, B.; Zhang, Y.; Gevorgian, V.; Denholm, P.; Hodge, B.M.; Hannegan, B. Achieving a 100% renewable grid: Operating electric power systems with extremely high levels of variable renewable energy. *IEEE Power Energy Mag.* **2017**, *15*, 61–73.
- Clegg, S.; Mancarella, P. Storing renewables in the gas network: Modelling of power-to-gas seasonal storage flexibility in low-carbon power systems. *IET Gener. Transm. Distrib.* **2016**, *10*, 566–575.
- Götz, M.; Lefebvre, J.; Mörs, F.; Koch, A.M.; Graf, F.; Bajohr, S.; Reimert, R.; Kolb, T. Renewable Power-to-Gas: A technological and economic review. *Renew. Energy* **2016**, *85*, 1371–1390.
- Zhu, R.; Jiang, J.; Sun, J.; Guo, D.; Wei, M.; Zhao, Z. Optimal planning of integrated electricity-natural gas distribution systems with hydrogen enriched compressed natural gas operation. *Int. J. Electr. Power Energy Syst.* **2024**, *161*, 110174.
- Davis, S.J.; Lewis, N.S.; Shaner, M.; Aggarwal, S.; Arent, D.; Azevedo, I.L.; Benson, S.M.; Bradley, T.; Brouwer, J.; Chiang, T.-M.; et al. Net-zero emissions energy systems. *Science* **2018**, *360*, eaas9793.
- Walawalkar, R.; Apt, J.; Mancini, R. Economics of electric energy storage for energy arbitrage and regulation in New York. *Energy Policy* **2007**, *35*, 2558–2568.
- Liu, X.; Wang, Y.; Wei, Z.; Tian, J.; Xiao, G.; Wang, P. Cost and capacity optimization of regional wind-hydrogen integrated energy system. *Int. J. Hydrogen Energy* **2024**, *49*, 571–585.
- Pan, G.; Gu, W.; Lu, Y.; Qiu, H.; Lu, S.; Yao, S. Optimal Planning for Electricity-Hydrogen Integrated Energy System Considering Power to Hydrogen and Heat and Seasonal Storage. *IEEE Trans. Sustain. Energy* **2020**, *11*, 2662–2676.
- Khodaei, A. Provisional microgrid planning. *IEEE Trans. Smart Grid* **2017**, *8*, 1096–1104.
- Zeng, B.; Zhao, L. Solving two-stage robust optimization problems using a column-and-constraint generation method. *Oper. Res. Lett.* **2013**, *41*, 457–461.
- Bertsimas, D.; Litvinov, E.; Sun, X.A.; Zhao, J.; Zheng, T. Adaptive robust optimization for the security constrained unit commitment problem. *IEEE Trans. Power Syst.* **2013**, *28*, 52–63.
- Wang, C.; Liu, F.; Wang, J.; Wei, W.; Mei, S. Risk-Based Admissibility Assessment of Wind Generation Integrated into a Bulk Power System. *IEEE Trans. Sustain. Energy* **2016**, *7*, 325–336.
- Liang, Z.; Chung, C.Y.; Wang, Q.; Chen, H.; Yang, H.; Wu, C. Fortifying renewable-dominant hybrid microgrids: A bi-directional converter based interconnection planning approach. *Engineering* **2025**, *51*, 130–143.
- Gurobi Optimization, LLC. Gurobi optimizer reference manual. 2023. Available online: <https://www.gurobi.com> (accessed on 3 April 2026).
- Chen, X.; Dong, W.; Yang, Q. Robust optimal capacity planning of grid-connected microgrid considering energy management under multi-dimensional uncertainties. *Appl. Energy* **2022**, *323*, 119642.
- Vielma, J.P. Mixed integer linear programming formulation techniques. *SIAM Rev.* **2015**, *57*, 3–57.
- Belotti, P.; Kirches, C.; Leyffer, S.; Linderoth, J.; Luedtke, J.; Mahajan, A. Mixed-integer nonlinear optimization. *Acta Numer.* **2013**, *22*, 1–131.
- Khan, T.; Yu, M.; Waseem, M. Review on recent optimization strategies for hybrid renewable energy system with hydrogen technologies: State of the art, trends and future directions. *Int. J. Hydrogen Energy* **2022**, *47*, 25155–25201.
- Kennedy, J.; Eberhart, R. Particle swarm optimization. In Proceedings of the ICNN'95—International Conference on Neural Networks, Perth, WA, Australia, 6 August 2002; Volume 4, pp. 1942–1948.
- Mirjalili, S.; Mirjalili, S.M.; Lewis, A. Grey Wolf Optimizer. *Adv. Eng. Softw.* **2014**, *69*, 46–61.
- Kaabeche, A.; Belhamel, M.; Ibtouen, R. Sizing optimization of grid-independent hybrid photovoltaic/wind power generation system. *Energy* **2011**, *36*, 1214–1222.
- Audet, C.; Hare, W. *Derivative-Free and Blackbox Optimization*; Springer: Berlin/Heidelberg, Germany, 2017.
- Schmidt, O.; Gambhir, A.; Staffell, I.; Hawkes, A.; Nelson, J.; Few, S. Future cost and performance of water electrolysis: An expert elicitation study. *Int. J. Hydrogen Energy* **2017**, *42*, 30470–30492.
- Hu, D.; Fan, Y.; Shao, W. Optimal sizing and operation of microgrid considering renewable energy uncertainty based on scenario generation. *J. Energy Storage* **2025**, *109*, 115174.
- U.S. Department of Energy; WINDEXchange. Wind Energy End-of-Service Guide. 2025. Available Online: <https://www.energy.gov/cmei/wind/windexchange/windexchange/wind-energy-end-service-guide> (accessed on 27 April 2026).
- Morales, H.; Conejo, A.J.; Liu, K.; Zhong, J. Pricing electricity in pools with wind producers. *IEEE Trans. Power Syst.* **2012**, *27*, 1366–1376.

30. Shahidehpour, M.; Fu, Y. Benders decomposition. *IEEE Power Energy Mag.* **2005**, *3*, 20–25.
31. Zhong, C.; Li, G.; Meng, Z.; Li, H.; Yildiz, A.R.; Mirjalili, S. Starfish optimization algorithm (SFOA): A bio-inspired metaheuristic algorithm for global optimization compared with 100 optimizers. *Neural Comput. Appl.* **2025**, *37*, 3641–3683.

**Disclaimer/Publisher’s Note:** The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.